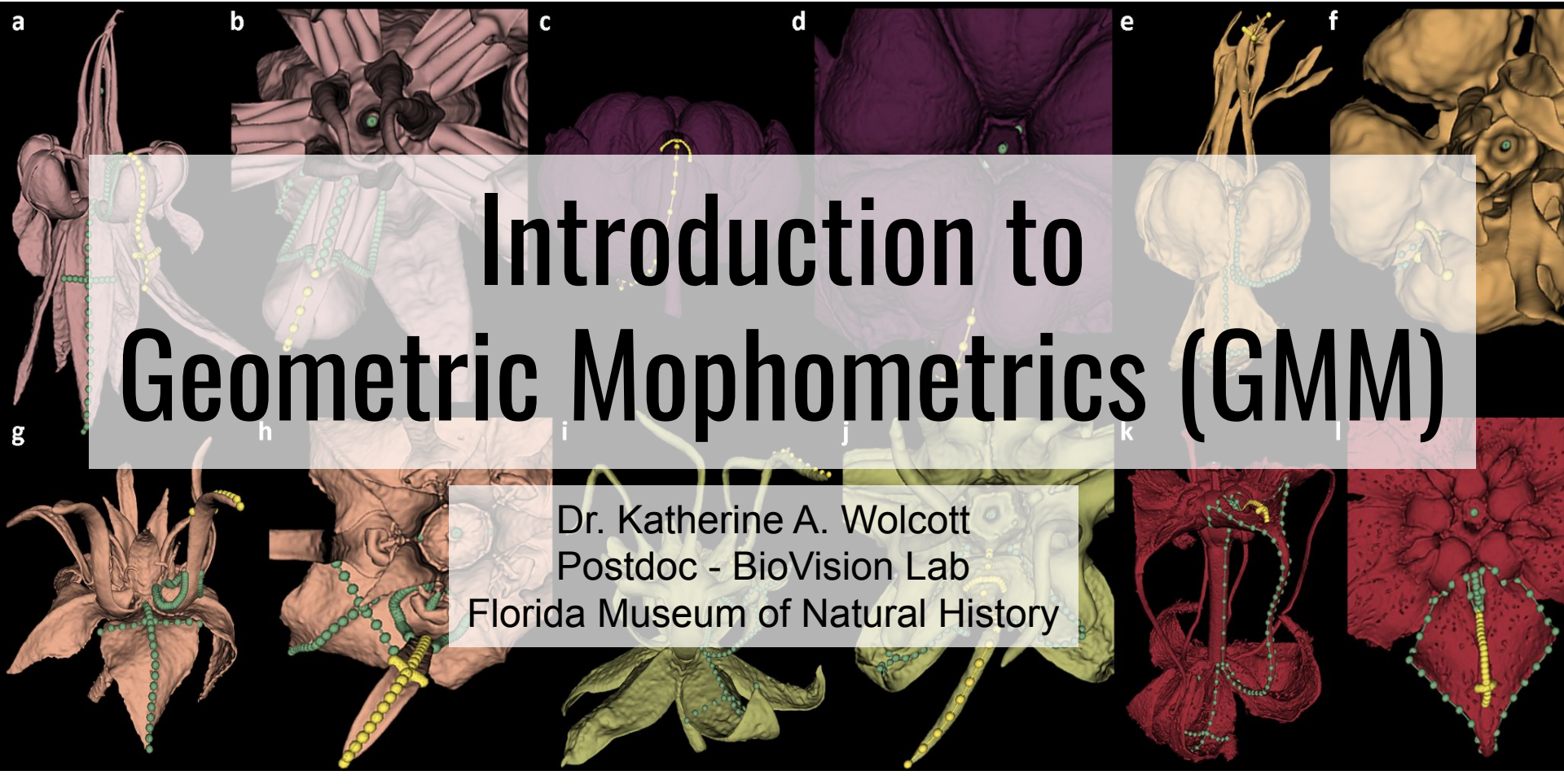
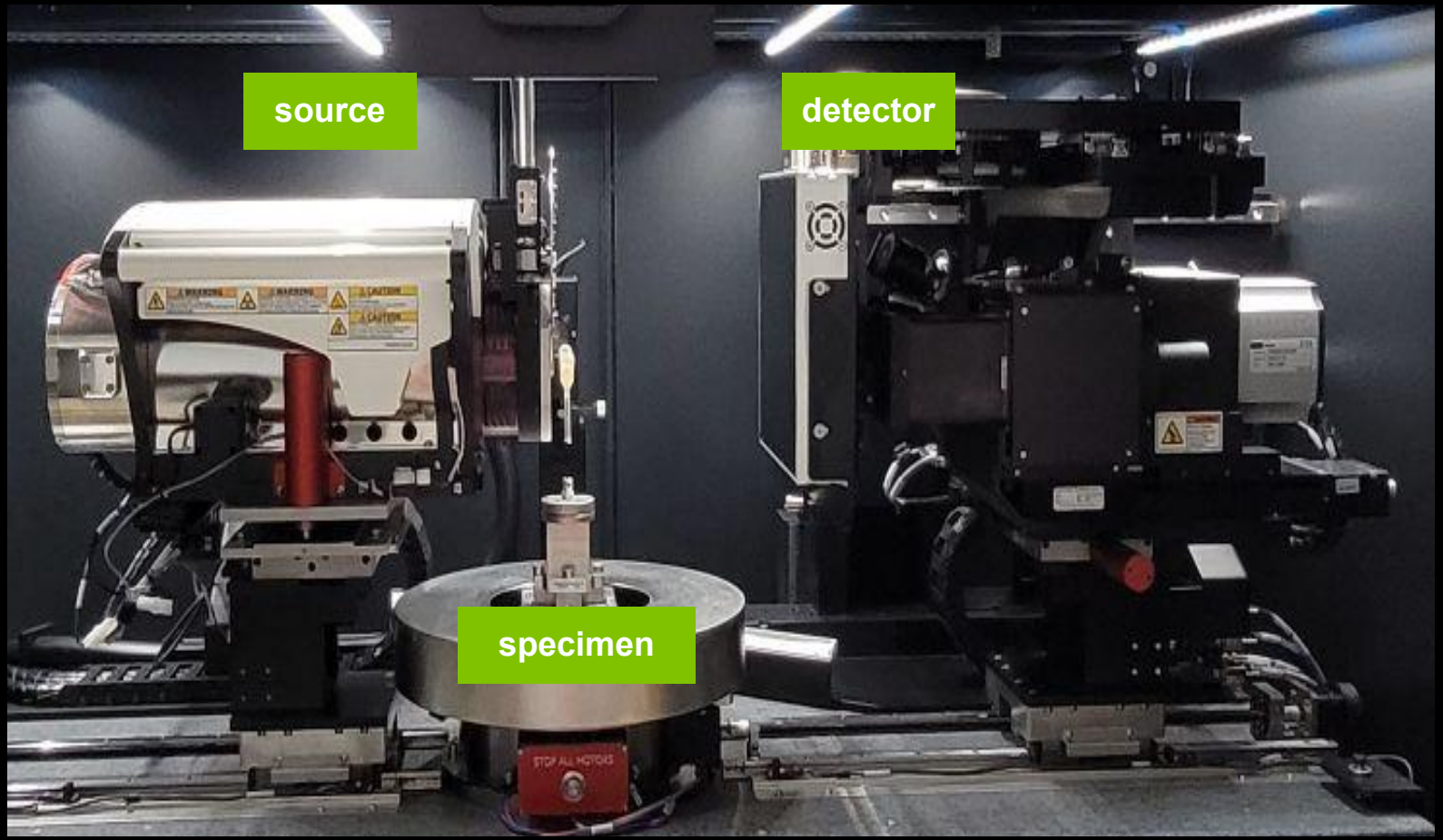


Introduction to Geometric Mophometrics (GMM)

Dr. Katherine A. Wolcott
Postdoc - BioVision Lab
Florida Museum of Natural History

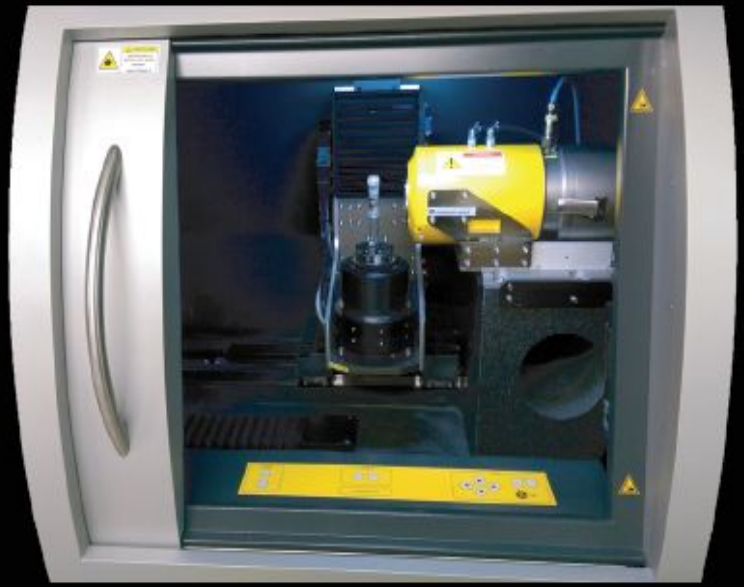
Introduction to micro-CT



From specimen to micro-CT data



contrast-enhanced
fixed specimen



mirco-CT scan



3D volume of micro-CT
dataset

From micro-CT data to mesh



micro-CT volume

Raw dataset
8-40GB
Requires special software



segmentation of parts

Intermediate annotated images



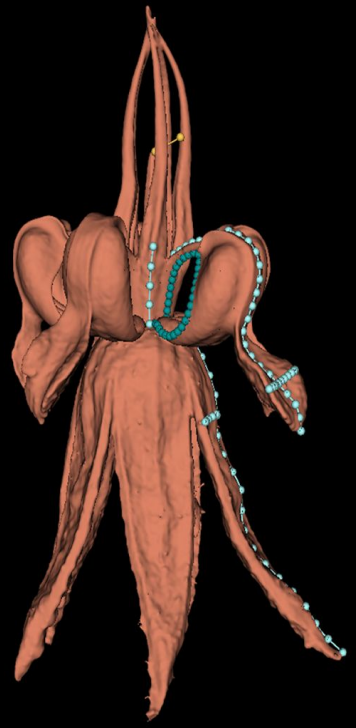
production of surfaces

Processed dataset
5-50MB
Can open on any computer

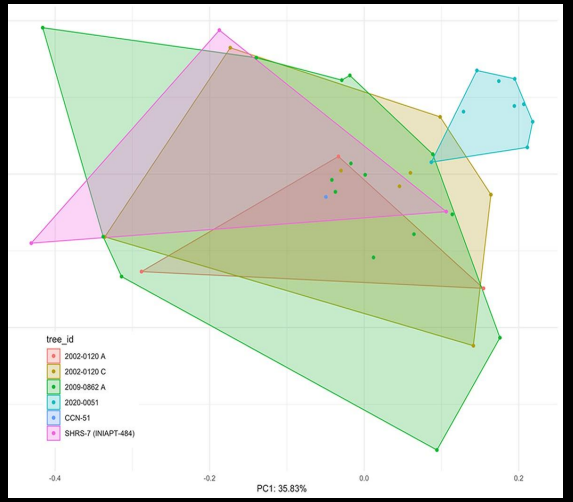
From mesh to GMM analysis



Surface Model



Landmarks



GMM Analysis

GMM use case: 3D pollination biology of cacao



Objectives

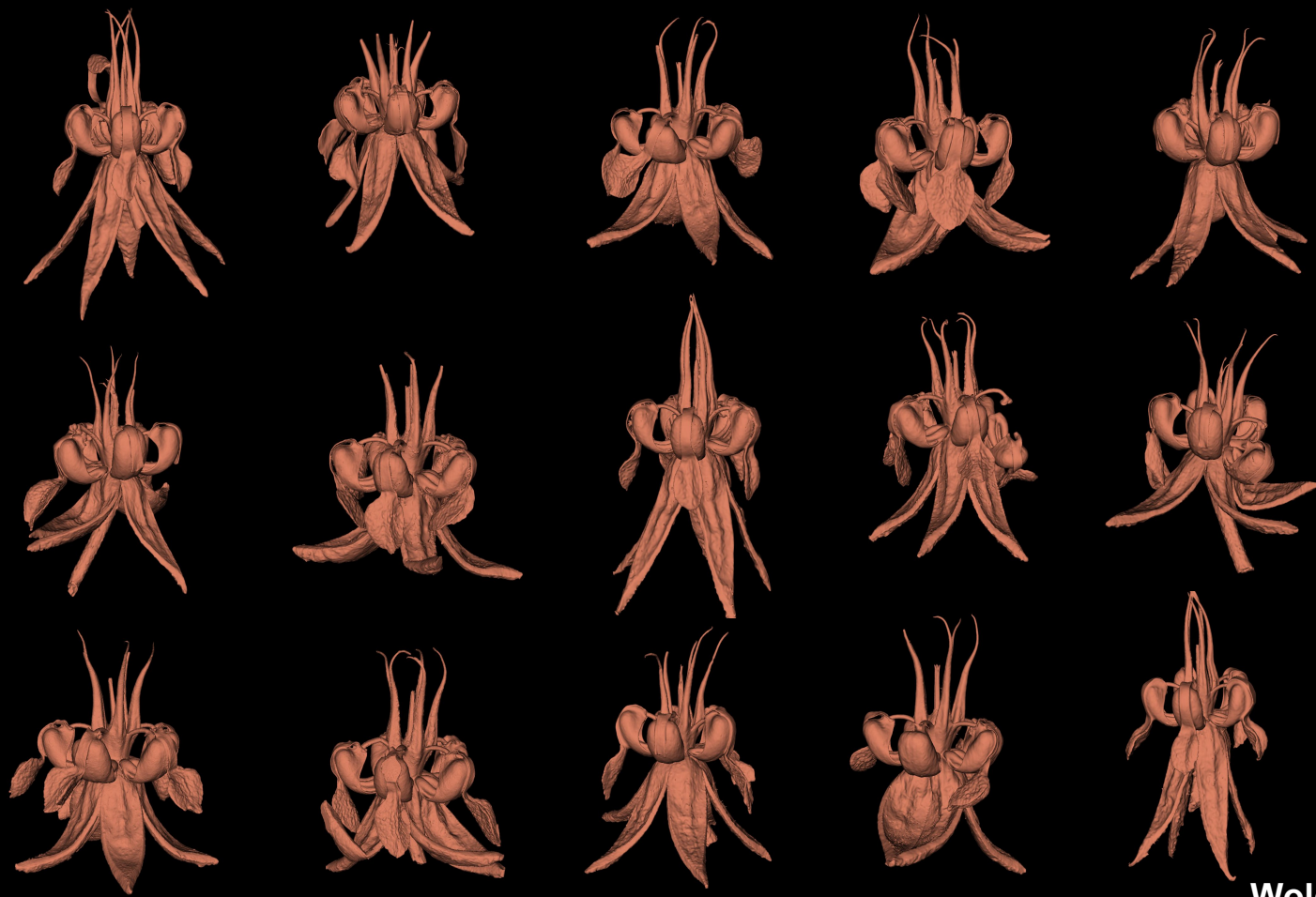
1. Obtain 50 micro-CT datasets of cacao and 20 of its hypothesized pollinator taxa

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator

3. Apply method to cacao and identify pollinators that fit and do not fit within the determined size limits

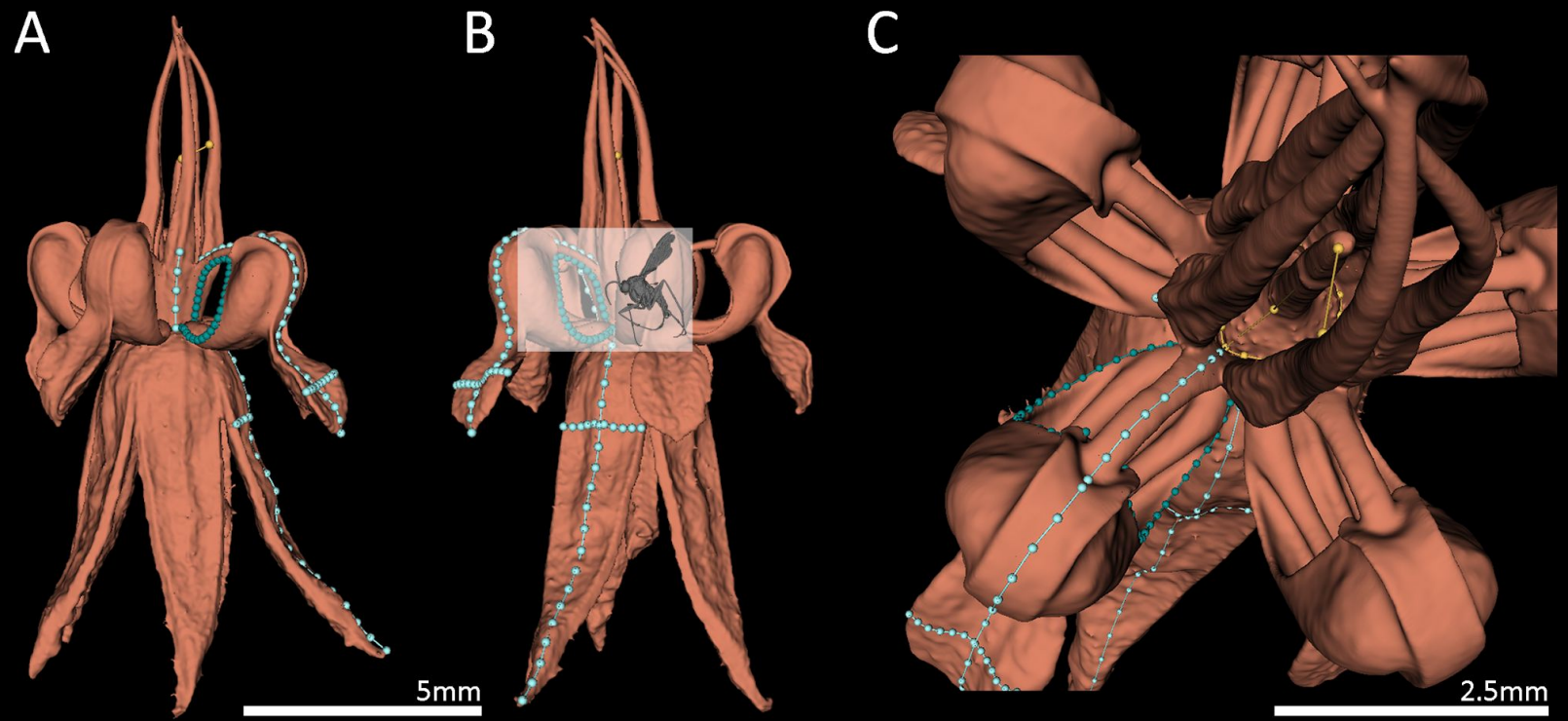
Wolcott et al. 2023

1. Obtain 50 micro-CT datasets of cacao and 20 of its hypothesized pollinator taxa



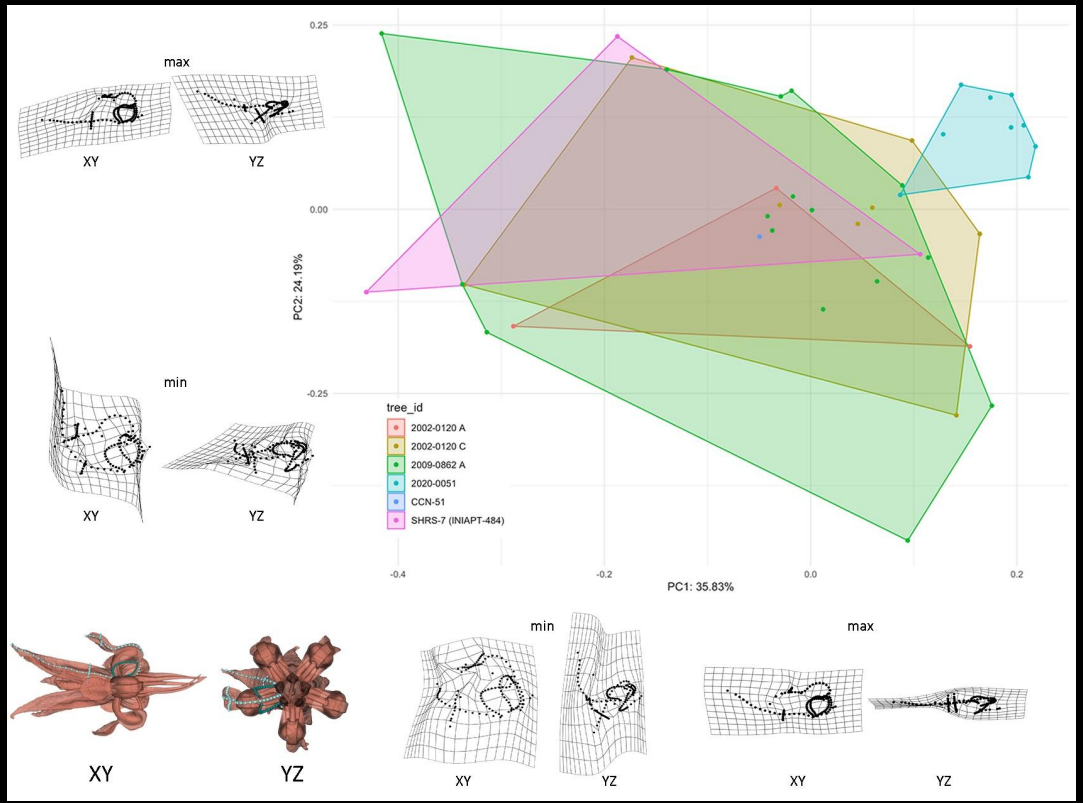
Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator



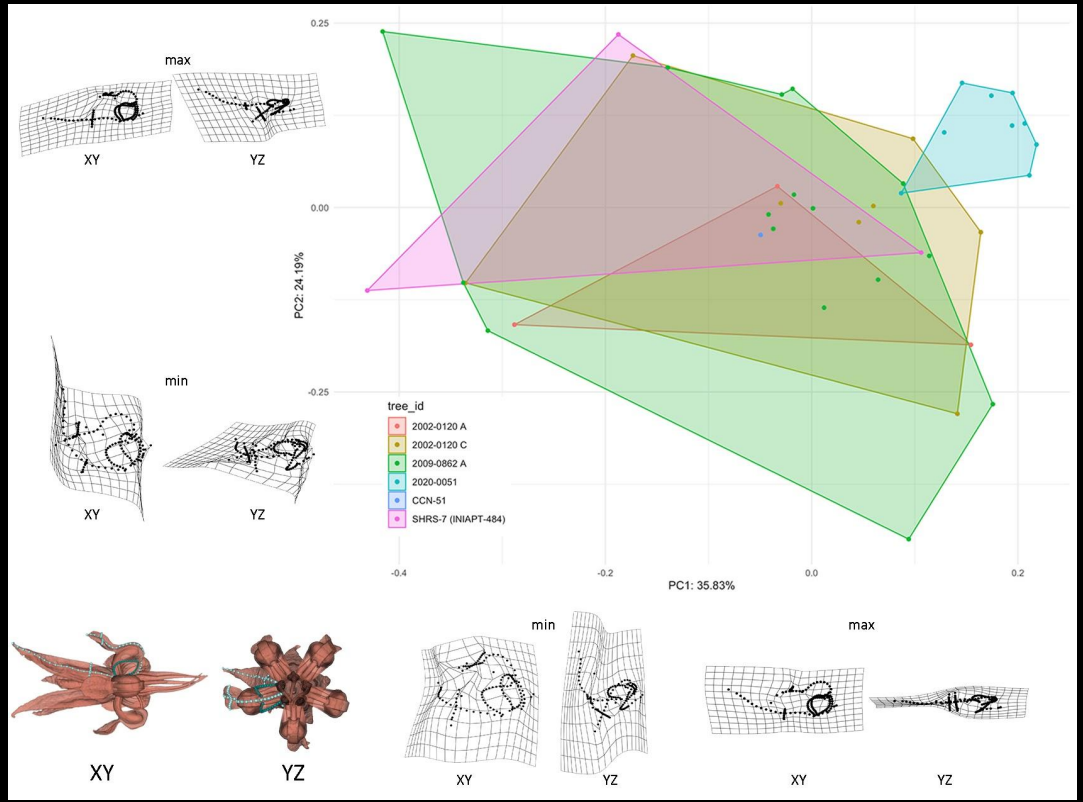
Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator



Wolcott et al. 2023

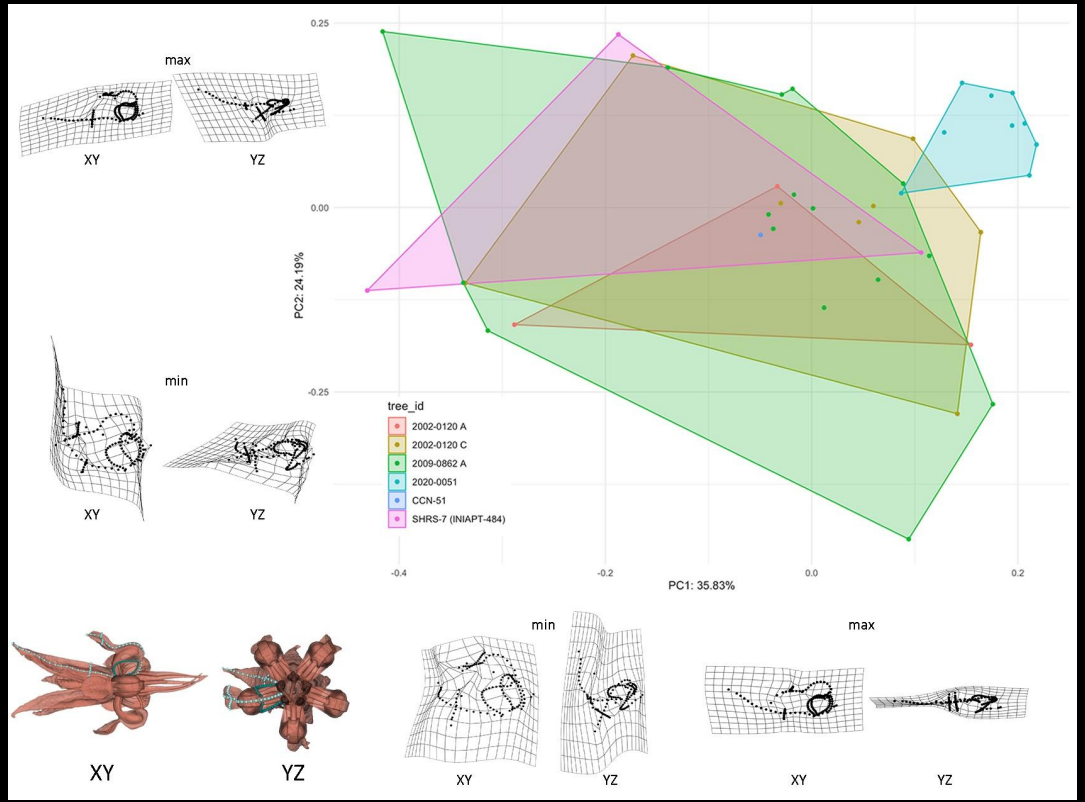
2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator



How can we quantify group effects and interactions?

Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator



How can we quantify group effects and interactions?

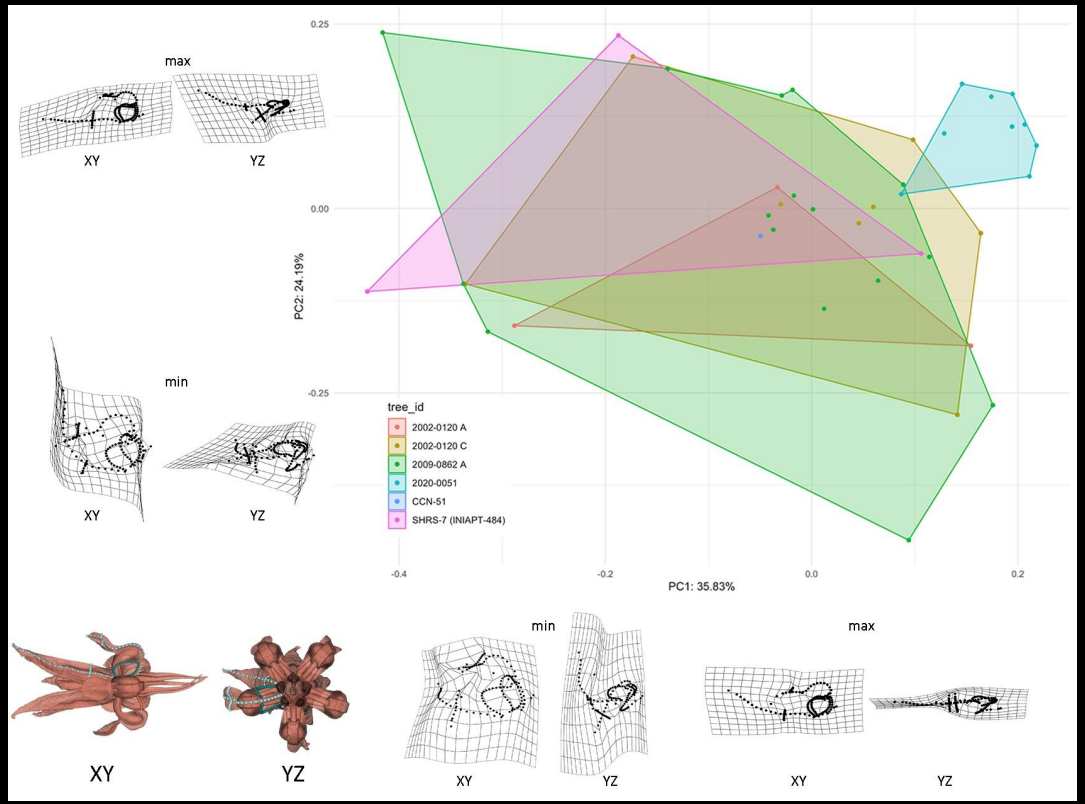
1) Are trees similar enough to treat as one group?

If yes, ask questions for all samples

If no, treat each tree separately

Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator



How can we quantify group effects and interactions?

1) Are trees similar enough to treat as one group?

If yes, ask questions for all samples

If no, treat each tree separately

Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator

Call: procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	0.900	423.00	0.001	***
Residuals	47	0.100			
Total	48				

ALL

How can we quantify group effects and interactions?

2) Can size alone explain all differences in shape? (allometry)

Null model

See example values

Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator

Call: procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	0.900	423.00	0.001	***
Residuals	47	0.100			
Total	48				

ALL

Df — Degrees of freedom

of independent pieces of information a term uses

Continuous predictor (size) = 1

Factor with k groups = k − 1

Total = sample size − 1

Residuals = what's left over to estimate error

How can we quantify group effects and interactions?

2) Can size alone explain all differences in shape? (allometry)

Wolcott et al. 2023

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ALL

How can we quantify group effects and interactions?

2) Can size alone explain all differences in shape? (allometry)

Rsq — R squared

Proportion of total shape variation a term explains

Term SS ÷ Total SS

Ranges 0 to 1 (0.125 = 12.5%)

All rows, including Residuals, sum to 1

Residuals = the share nothing in the model explains

Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator

Call: procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	0.900	423.00	0.001	***
Residuals	47	0.100			
Total	48				

ALL

F — F value

Signal to noise, adjusted for Df

(Term SS ÷ Term Df) ÷ (Residual SS ÷ Residual Df)

Larger F: stronger effect relative to error

F ≈ 1 : term explains no more than noise

How can we quantify group effects and interactions?

2) Can size alone explain all differences in shape? (allometry)

Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator

Call: procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	0.900	423.00	0.001	***
Residuals	47	0.100			
Total	48				

ALL

How can we quantify group effects and interactions?

2) Can size alone explain all differences in shape? (allometry)

P — P value

How often chance would produce F

Computed by permutation (eg, n=1000)

Shuffle residuals 1000× : How often does chance beat F?

P = 0.001 : essentially never

Smaller P: less likely the pattern is chance

Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator

Call: procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	0.900	423.00	0.001	***
Residuals	47	0.100			
Total	48				

ALL

	Df	Rsq	F	P	
log(Csize)	1	0.125	6.72	0.001	***
Residuals	47	0.875			
Total	48				

SOME

How can we quantify group effects and interactions?

2) Can size alone explain all differences in shape? (allometry)

Null model

See example values

Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator

Call: procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	0.125	6.72	0.001	***
Residuals	47	0.875			
Total	48				

SOME

How can we quantify group effects and interactions?

2) Can size alone explain all differences in shape? (allometry)

Null model

See actual values

Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator

Call: procD.lm(f1 = coords ~ Csize + self_compat)

	Df	Rsq	F	P	
Csize	1	0.110	6.29	0.001	***
self_compat	1	0.086	4.92	0.001	***
Residuals	46	0.804			
Total	48				

Yes (?)

How can we quantify group effects and interactions?

3) Allometry + group effects

Does shape differ between groups once size is accounted for?

Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator

Call: procD.lm(f1 = coords ~ Csize * self_compat)

	Df	Rsq	F	P	
Csize	1	0.110	6.35	0.001	***
self_compat	1	0.086	4.97	0.001	***
Csize:self_compat	1	0.025	1.44	0.213	
Residuals	45	0.779			
Total	48				

NO

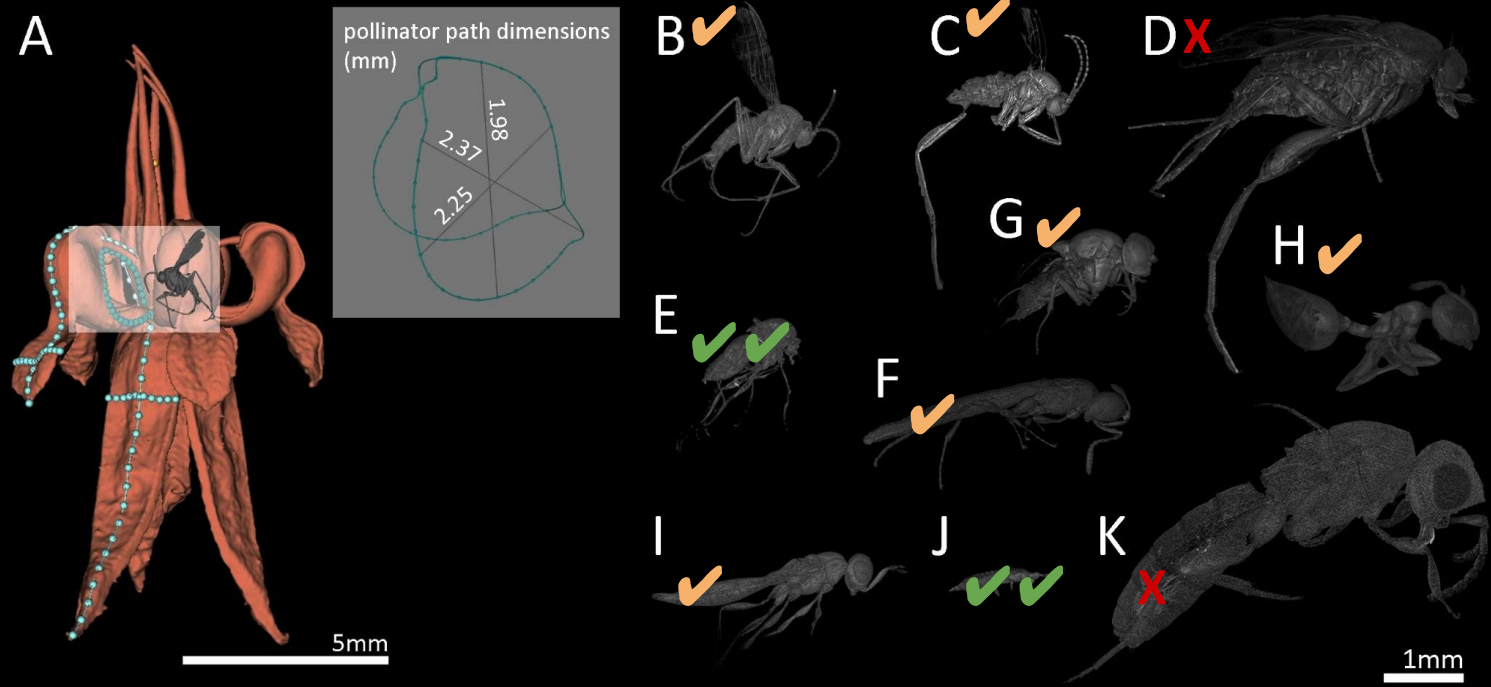
How can we quantify group effects and interactions?

4) Allometry * group effects

Do groups differ in how shape changes with size?

Wolcott et al. 2023

3. Apply method to cacao and identify pollinators that fit and do not fit within the determined size limits



Wolcott et al. 2023

GMM use case: 3D pollination biology of cacao



Objectives

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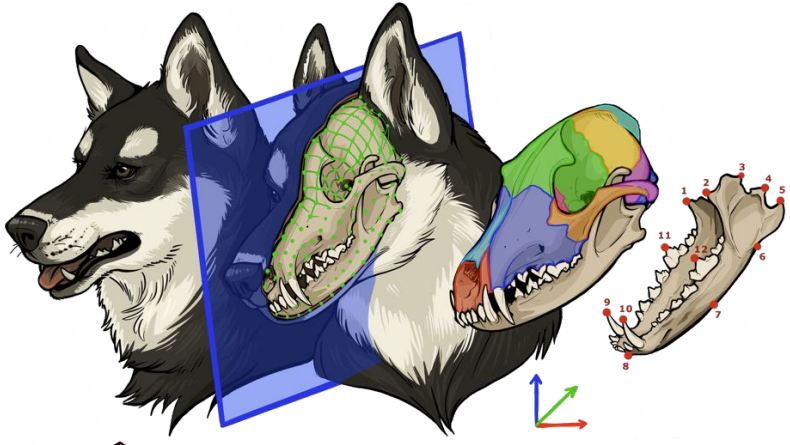
Wolcott et al. 2023



Questions?

How to do your own 3D GMM analysis?

GEOMORPH



SLICERMORPH

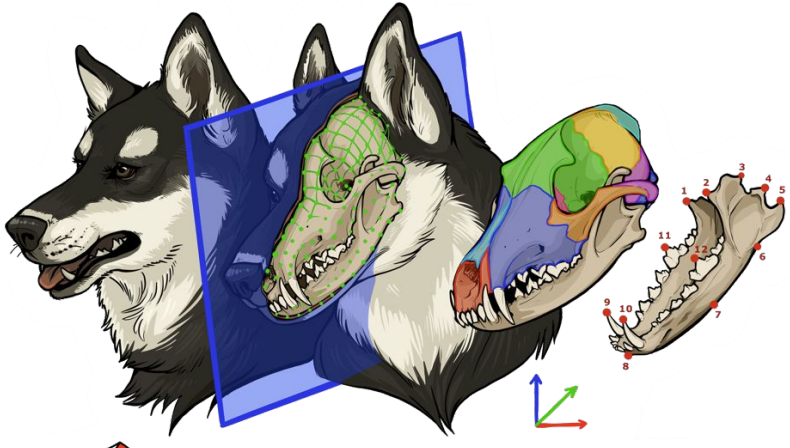
VISUALIZE • SEGMENT • DIGITIZE • ANALYZE • SHARE

Adams et al. 2025

Rolfe et al. 2021

How to do your own 3D GMM analysis?

GEOMORPH



SLICERMORPH

VISUALIZE • SEGMENT • DIGITIZE • ANALYZE • SHARE

Workshop Materials on GitHub

SlicerMorph Tutorial
Demo Data - Lab Mice Skulls, LMs, (fake) metadata
Google Colab Geomorph demo